

Quantile MDP with Markovian Policy reduces to Quantile-based Distributional RL

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1 Notation and General Assumptions

Denote the random variable by a tilde, i.e., $\tilde{x}$. Denote the augmented real as $\bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$.

Quantiles and Value-at-Risk. The quantile of random variable $\tilde{x}$ at level $\alpha \in [0, 1]$ is any $\tau \in \mathbb{R}$ such that $\mathbb{P}[\tilde{x} \leq \tau] \geq \alpha$ and $\mathbb{P}[\tilde{x} < \tau] \geq 1 - \alpha$. It might not be unique and lies in the interval $[q_\alpha^-(\tilde{x}), q_\alpha^+(\tilde{x})]$, where

$$\begin{aligned} q_\alpha^-(\tilde{x}) &:= \min\{\tau \in \bar{\mathbb{R}} | \mathbb{P}[\tilde{x} \leq \tau] \geq \alpha\} \\ q_\alpha^+(\tilde{x}) &:= \max\{\tau \in \bar{\mathbb{R}} | \mathbb{P}[\tilde{x} < \tau] \leq \alpha\}. \end{aligned}$$

$\text{VaR}_\alpha[\tilde{x}]$ is defined as the largest $1 - \alpha$ confidence lower bound on the value of $\tilde{x}$, i.e.,

$$\text{VaR}_\alpha[\tilde{x}] := q_\alpha^+(\tilde{x}).$$

For ease of presentation, we assume the quantile is unique, i.e., $q_\alpha^-(\tilde{x}) = q_\alpha^+(\tilde{x})$, for any $\tilde{x}$. In this case, value-at-risk (VaR) equals quantile.

For a Markov Decision Process (MDP), we also assume the space of state, action, and random reward is finite.

2 Quantile-based Distributional RL

Distributional reinforcement learning is getting popular since the seminal work of [Bellemare et al., 2017], where a distributional Bellman equation is proposed, i.e.,

$$\tilde{z}(s, a) \stackrel{d}{=} \tilde{r}(s, a) + \gamma \tilde{z}(s', \tilde{a}') \quad (1)$$

In this equation, $\tilde{z}(s, a) := \sum_{t=0}^{\infty} [\gamma^t \tilde{r}(\tilde{s}_t, \tilde{a}_t) | \tilde{s}_0 = s, \tilde{a}_t = a]$ is the random variable of the return, and $\stackrel{d}{=}$ means equal in distribution. [Bellemare et al., 2017] mentioned that when $\tilde{a}'$ comes from a fixed policy $\pi(\cdot | s')$, i.e., performing policy evaluation, the equation enjoys the contraction property. However, in control setting, i.e., $\tilde{a}'$ chooses the optimal risk-neutral action, the distributions of two sides of Eq. 1 might not be equal.

[Bellemare et al., 2017] used categorical distribution to represent the value distribution. Later, [Dabney et al., 2018] proposed to represent the value distribution by its empirical inverse CDF (quantile function), and update the quantile estimates by quantile regression given sampled (s, a, r, s') as

$$q_\alpha(s, a) \leftarrow q_\alpha(s, a) - \eta \cdot \partial_y \mathbb{E}_{u \sim U[0,1]} \left[l_\alpha(r + \gamma q_u(s', a') - y) \right] \Big|_{y=q_\alpha(s,a)}, \quad \forall \alpha \in [0, 1] \quad (2)$$

where $q_\alpha(s, a) := \text{VaR}_\alpha[\tilde{z}(s, a)]$ represents the α quantile of the state-action value; η is the learning rate; $U[0, 1]$ is a uniform distribution on $[0, 1]$; $l_\alpha(\cdot)$ is the loss function corresponds to quantile regression given by

$$l_\alpha(x - y) := (\alpha - \mathbb{I}\{x < y\})(x - y);$$

a' can come from $\pi(\cdot | s')$ if doing policy evaluation, and $a' = \arg \max_a \mathbb{E}_u[q_u(s', a)]$ when performing optimal control.

Quantile-based approach has become one of the mainstream approaches of distributional RL since it was introduced. Recently, [Rowland et al., 2024] provided theoretical analysis for quantile TD learning (Eq. 2).

3 Quantile MDP

Consider a more general problem, given an MDP, what is the optimal quantile value that can be achieved in the MDP for all $\alpha \in [0, 1]$ and (s, a) , i.e.,

$$q_\alpha^*(s, a) := \max_{\pi_h} \text{VaR}_\alpha^{\pi_h} \left[\sum_{t=0}^{\infty} \gamma^t \tilde{r}(\tilde{s}_t, \tilde{a}_t) \mid \tilde{s}_0 = s, \tilde{a}_0 = a \right]. \quad (3)$$

The corresponding optimal policy of Eq. 3 is history-dependent since VaR_α operator is non-linear, therefore denoted as π_h (h means history) in Eq. 3.

[Li et al., 2022] and [Hau et al., 2023] showed that this static VaR problem enjoys a dynamic decomposition which admits a Bellman-like equation. However, the equations in [Li et al., 2022] and [Hau et al., 2023] are model-based where a constrained optimization problem involving the transition probability need to be solved. To address this limitation, [Hau et al., 2025] proposed a nested VaR Bellman optimality equation as¹

$$q^*(s, \alpha, a) = \text{VaR}_\alpha \left[\tilde{r}(s, a) + \gamma \max_{a'} q^*(\tilde{s}', \tilde{u}, a') \right], \quad (4)$$

where $\tilde{u}$ follows $U[0, 1]$ and VaR_α is applied to the joint distribution of $\tilde{r}(s, a)$, $\tilde{s}'$ and $\tilde{u}$. We have moved the quantile-level α inside the q function to treat it as an augmentation of the state space. The fixed point of Eq. 4 is the optimal α -quantile under (s, a) pair, i.e., Eq. 3.

Eq. 4 is not directly amenable to a Q -learning style algorithm, since the VaR operator is generally unavailable in closed form. However, notice that α -quantile is the argmin of the quantile regression loss, as a result, Eq. 4 can be alternatively expressed as

$$q^*(s, \alpha, a) = \arg \min_y \mathbb{E} \left[l_\alpha \left(\tilde{r}(s, a) + \gamma \max_{a'} q^*(\tilde{s}', \tilde{u}, a') - y \right) \right]. \quad (5)$$

As a result, the quantile value can be updated, by gradient descent, towards the decreasing direction of the quantile regression loss. This can be performed by stochastic gradients given sampled (s, α, a, r, s') (Note that using sampled r and s' still gives unbiased gradient). The quantile value is updated by

$$q(s, \alpha, a) \leftarrow q(s, \alpha, a) - \eta \cdot \partial_y \mathbb{E}_{u \sim U[0, 1]} \left[l_\alpha \left(r + \gamma \max_{a'} q(s', u, a') - y \right) \right] \Big|_{y=q(s, \alpha, a)}, \quad (6)$$

where η is the learning rate.

Eq. 2 and Eq. 6 share very similar structure. The only difference is that Eq. 6 obtains the optimal action a' for each quantile-level $u \in [0, 1]$, while Eq. 2, in control setting, uses the optimal risk-neutral action $a' = \arg \max_a \mathbb{E}_u [q(s', u, a)]$. However, this subtle difference suggests that quantile-based distributional RL approach, in control setting, is **not** learning the optimal quantile values in an MDP. Another explanation is that the optimal quantile policy is history-dependent while distributional RL uses Markovian policy, see Algo. 1 of [Hau et al., 2025] on how to obtain the optimal history-dependent VaR policy.

¹See the justification of including random reward $\tilde{r}$ in this Bellman equation in [Luo and Delage, 2026].

4 Quantile Decomposition under Markovian Policy

We show that when policy is Markovian, the quantile decomposition technique (Li et al. [2022]) used in quantile MDP reduces to quantile-based distributional RL approach in policy evaluation.

$$\begin{aligned}
q^\pi(s, \alpha, a) &= \text{VaR}_\alpha \left[\sum_{t=0}^{\infty} \gamma^t \tilde{r}(\tilde{s}_t, \tilde{a}_t) \mid \tilde{s}_0 = s, \tilde{a}_0 = a \right] \\
&= \text{VaR}_\alpha \left[\tilde{r}(s, a) + \gamma \sum_{t=1}^{\infty} \gamma^{t-1} \tilde{r}(\tilde{s}_t, \tilde{a}_t) \right], \quad \tilde{a}_t \sim \pi(\cdot \mid \tilde{s}_t) \\
&\stackrel{(a)}{=} \text{VaR}_\alpha \left[\sum_{(r_i, s'_i, a'_i)} \pi(a'_i \mid s'_i) P(r_i, s'_i \mid s, a) \left(r_i + \gamma \left[\sum_{t=1}^{\infty} \gamma^{t-1} \tilde{r}(\tilde{s}_t, \tilde{a}_t) \mid \tilde{s}_1 = s'_i, \tilde{a}_1 = a'_i \right] \right) \right], \quad \tilde{a}_t \sim \pi(\cdot \mid \tilde{s}_t) \\
&\stackrel{(b)}{=} \text{VaR}_\alpha \left[\sum_{(r_i, s'_i, a'_i)} P(r_i, s'_i, a'_i \mid s, a) \left(r_i + \gamma q^\pi(s'_i, \tilde{u}, a'_i) \right) \right] \\
&\stackrel{(c)}{=} \max_{\beta} \min_i \text{VaR}_{\beta_i} [r_i + \gamma q^\pi(s'_i, \tilde{u}, a'_i)] \quad \text{with} \quad \sum_i \beta_i \cdot P(r_i, s'_i, a'_i \mid s, a) \leq \alpha \\
&= \max_{\beta} \min_i r_i + \gamma q^\pi(s'_i, \beta_i, a'_i) \\
&\stackrel{(d)}{=} \text{VaR}_\alpha \left[\tilde{r} + \gamma q^\pi(\tilde{s}', \tilde{u}, \tilde{a}') \right] \quad \text{with} \quad \tilde{a}' \sim \pi(\cdot \mid \tilde{s}'),
\end{aligned}$$

where (a) considers all the possible combinations of (r, s', a') (indexed by i) under the joint distribution of policy $\pi(a' \mid s')$ and transition $P(r, s' \mid s, a)$; (b) replaces $\pi(a'_i \mid s'_i) P(r_i, s'_i \mid s, a)$ by $P(r_i, s'_i, a'_i \mid s, a)$ and replaces the distribution of $\left[\sum_{t=1}^{\infty} \gamma^{t-1} \tilde{r}(\tilde{s}_t, \tilde{a}_t) \mid \tilde{s}_1 = s'_i, \tilde{a}_1 = a'_i \right]$ by its equivalent representation $q^\pi(s'_i, \tilde{u}, a'_i)$; (c) is according to the quantile decomposition theory, i.e., theorem 1 and Lemma 2 in [Li et al., 2022]; (d) is according to Lemma B.4 of [Hau et al., 2025].

Following the same idea as Eq. 5 and Eq. 6, we have

$$\begin{aligned}
q^\pi(s, \alpha, a) &= \text{VaR}_\alpha \left[\tilde{r}(s, a) + \gamma q^\pi(\tilde{s}', \tilde{u}, \tilde{a}') \right] \\
&= \arg \min_y \mathbb{E} \left[l_\alpha(\tilde{r}(s, a) + \gamma q^\pi(\tilde{s}', \tilde{u}, \tilde{a}') - y) \right],
\end{aligned}$$

which leads to the update rule as Eq. 2 given sampled (s, a, r, s') when performing policy evaluation. Therefore, quantile MDP with Markovian policy reduces to quantile-based distributional RL under policy evaluation.

5 Remark

Note that some technique details are omitted. Please refer to [Hau et al., 2025] for handling the non-smoothness of quantile regression loss, and the convergence analysis for Q -learning algorithm in quantile MDP.

References

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